

Module 6.2: Security Services

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Agenda



- ▶ Traffic Control with ACLs
- ▶ ACL Features
- ▶ SNMP
- ▶ NetFlow
- ▶ Port Mirroring
- ▶ Syslog Servers
- ▶ NTP
- ▶ AAA Servers
- ▶ VPN
- ▶ Summary & Key Takeaways

Traffic Control with ACLs

- ▶ ACLs (Access Control Lists) control traffic flow.
- ▶ They permit or deny traffic based on rules.
- ▶ Used on routers and firewalls.

ACLs: Important Features

- ▶ Standard ACLs filter by source IP.
- ▶ Extended ACLs filter by IP, protocol, and ports.
- ▶ Rules are processed top-down.

SNMP

- ▶ SNMP (Simple Network Management Protocol) monitors devices.
- ▶ Allows administrators to collect statistics.
- ▶ Used for performance and fault monitoring.

NetFlow



- ▶ NetFlow collects traffic flow data.
- ▶ Helps detect abnormal behavior.
- ▶ Used for visibility and security analysis.

Port Mirroring

- ▶ Copies traffic to monitoring tools.
- ▶ Used for analysis and troubleshooting.
- ▶ Does not disrupt live traffic.

Syslog Servers

- ▶ Syslog collects device log messages.
- ▶ Helps detect incidents and errors.
- ▶ Centralized logging improves visibility.

NTP

- ▶ NTP (Network Time Protocol) synchronizes time.
- ▶ Accurate time is critical for logs and forensics.
- ▶ Prevents time-based attacks.

AAA Servers

- ▶ AAA stands for Authentication, Authorization, Accounting.
- ▶ Controls user access.
- ▶ Tracks user activity.

VPN



- ▶ VPN (Virtual Private Network) encrypts traffic.
- ▶ Protects data over untrusted networks.
- ▶ Used for remote access and site-to-site connections.

Summary & Key Takeaways

- ▶ Security services support network protection.
- ▶ They improve control, monitoring, and visibility.
- ▶ Together they strengthen network security.